

Shucheng (Bangli) Cao

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CAREER OBJECTIVE**Primary:** Target Validation Scientist (Genetics-based) and Statistical Genetics Researcher**Open to:** Multi-omics data science, biopharma data analysis / data science, biopharma financial investment research (quant finance / hedge fund / PE / VC / investment banking), life sciences consulting (management & scientific)**PROFESSIONAL SUMMARY**

- **Healthcare data scientist (PhD-level)** with 5+ years analyzing large-scale real-world data (790K+ patients). Apply causal inference and ML to identify unmet medical needs; pharma sciences background means I know which data patterns actually matter clinically.
- **Google Scholar citations >1,000; H-index = 6.** 2 publications in top journals (IEEE TPAMI IF 24.3, Advanced Science IF 17.5).
- Leveraged AI tools (Claude, ChatGPT) for hypothesis generation, literature synthesis, and presentation — compressing a week-long project scoping and pitch cycle to under half a day (>700% efficiency gain).
- **CAS\$350,000+ in competitive funding:** KAUST MS Fellowship (~CA\$200,000, top ~0.1%); FRQNT B2X (~CA\$80,000, top ~0.1%, province-wide); McGill PhD Scholarship (~CA\$60,000, top 20%); McGill Medicine Internal Scholarship (top 5%); PGSS Travel Award (top 1%); QLS Travel Award (×3, top 15%).
- **Presented** to 500+ attendees at Guangdong Medical University (invited); poster at ASHG 2024; work reviewed by McGill department chair, dean, and cross-faculty committees; conducted under co-supervision of Dr. Brent Richards (CEO, 5 Prime Sciences), bridging academic research and biotech translation.
- **Outside the lab:** gym regular, amateur vocalist, philosophy reader (Buddhist, Daoist, and the practical kind), and someone who will always find the best local food on a work trip.

EDUCATION**McGill University**

PhD candidate in Quantitative Life Sciences

2022-present

Montreal

King Abdullah University of Science and Technology

Master of Science in Bioengineering

2020-2022

Jeddah

China Pharmaceutical University

Bachelor of Science in Pharmaceutical Analysis

2015-2019

Nanjing

WORK EXPERIENCE**Ph.D. Researcher (Supervisor: Dr. Guillaume Butler-Laporte, McGill University)****09/2022-present**

- **Problem statement:** *Obesity drives nonalcoholic fatty liver disease in >1B people with few approved drugs. Which proteins in the blood actually cause the disease, and can we drug them?*
 - **Hypothesized** that specific blood proteins are not just correlated with liver disease — they actually cause it.
 - **Analyzed genetic and protein data from >790,000 people** using causal inference method (Mendelian randomization) to separate real biological signals from statistical noise.
 - **Built a cross-validation pipeline** across two independent datasets to ensure reproducibility beyond platform-specific artifacts.
 - **Identified 4 new proteins** that may drive disease progression and evaluated whether they can realistically be targeted by drugs
- Impact:** Prioritize drug targets in a \$2B+ market with only ~5% historical drug success rate.
Achievement: a first-author manuscript (submitted); 10+ presentations; >CA\$150,000 in competitive scholarships.

Research Associate (Advisor: Prof. Charlotte Hauser, King Abdullah Univ of Sci and Tech)**09/2021-05/2022**

- **Problem statement:** *What makes peptides stick together and form structures? This matters because these structures are used as biomaterials for 3D printing and for studying diseases like Alzheimer's (55M patients, no cure).*
- **Hypothesized that** the driving force is how much a peptide repels water (hydrophobicity), not its ring-shaped chemical structure (aromaticity) — challenging the decade-old consensus.
- Ran head-to-head computational simulations to test both explanations; **Disproved the old model** and confirmed water-repulsion (hydrophobicity) as the key factor.
- **Characterized 6 novel ultrashort peptides** and built a ranking system based on this property — replacing years of lab trial-and-error with a predictive shortlist for biomaterial designers.

- **Showed computationally** that a drug candidate can block the protein clumping linked to Alzheimer's (55M patients, no cure) — **first proof-of-concept for this drug class** in simulation.

Impact: Corrected a 20+ year design assumption (Gazit, 2002); provided first computational proof-of-concept for blocking Alzheimer's-linked protein aggregation in a disease with \$3.8B+ in annual global research funding and no cure.

Achievement: Top 10 graduation thesis at KAUST; >CA\$200,000 scholarships; >5+ presentations.

International Collaboration (Advisor: Dr. Jun Ma, University of Toronto)

02/2020-07/2020

- **Problem statement:** *AI tools that read medical scans scored near-perfectly in lab tests — but no one had tested them across different hospitals. In the U.S. alone, misdiagnosis harms 800,000+ patients and kills 370,000+ per year.*
- **Hypothesized that these AI models only looked accurate** because all the test data came from single hospitals under ideal conditions — hiding failures that would show up in real-world use.
- **Co-built the largest multi-hospital CT scan dataset at the time** (1,112 scans, 12 hospitals) to test whether top AI models hold up outside the lab.
- **Tested top AI models on real hospital data** — found they failed badly, preventing premature use on real patients and pushing the field to prioritize real-world accuracy.
- **Released the dataset** and testing tools freely available online; now used by 4,000+ research teams worldwide.

Impact: Set the validation standard for a \$6B+ medical imaging AI industry before faulty tools could reach patients at scale.

Achievement: 1 publication — IEEE TPAMI (IF: 24.3); dataset adopted as a gold standard in medical imaging community.

Research Assistant (Advisor: Prof. Zheng Li, Weill Cornell Medicine)

09/2019-07/2020

- **Problem statement:** *Over 90% of cancer drugs fail in clinical trials — largely because flat cell tests in labs do not reflect how real tumors behave in the body.*
- **Hypothesized** that 3D lab-grown tumor models and humanized mice would predict drug response more accurately than standard flat-cell setups.
- **Built a 3D tumor-on-scaffold platform** that outperformed flat-cell screens at predicting which drugs work in humans.
- **Developed a humanized mouse model** to test an immune-checkpoint drug (BMS202) for colorectal cancer (~900K deaths/year).

Impact: Gave drug developers more reliable screening tools where each late-stage failure wastes ~\$1.3B and years of R&D.

Achievement: 2 peer-reviewed publications — Advanced Science (IF: 17.5), etc.

LEADERSHIP & COMMUNICATION

Event coordinator - Academic events (Lady Davis Institute Trainee Committee)

07/21/2025

- Organized the Lady Davis Institute Research Day (300+ attendees, 10+ institutions): led abstract review, set selection criteria, and coordinated cross-institutional logistics.
- Organized epidemiology round tables (50+ attendees, 3 institutions) featuring speakers from professor to postdoc to PhD student, giving trainees a window into research at every career stage.

Workshop lead (McGill Initiative in Computational Medicine)

02/2025-03/2026

- Designed and delivered comprehensive hands-on training sessions for graduate students across multiple topics, including Polygenic Risk Scores (PRS), Introduction to R, Statistics in R, and Linux.
- Developed structured teaching materials and coding exercises, while providing real-time coding support, debugging assistance, and statistical guidance.

SELECTED PUBLICATIONS

1. Shucheng Cao, et al. The impact of BMI-driven proteins on MASLD pathogenesis (Submitted).
2. Wanheng Zhang^a, Shucheng Cao, et al. *Advanced Science* (2023). (IF: 17.521) → [Link](#)
3. Cao, Shucheng. *Molecular Dynamics Simulations to Study the Mechanism of Self-Assembly of Amphiphilic*. Diss. 2022. → [Link](#)
4. Pengzhao Shang^a, Shucheng Cao, et al. *Acta Biochimica et Biophysica Sinica* (2022). (IF: 3.511) → [Link](#)
5. Jun Ma, Shucheng Cao, et al. (2021/07/27). *IEEE TPAMI* (IF: 24.314). → [Link](#)
6. Ma Jun, Cao Shucheng, et al. (2020/04/20). COVID-19 CT Dataset. Zenodo. → [Link](#)

SKILLS

- **Programming & Tools:** R · Python · Linux/Unix · Shell scripting
- **Problem-Solving:** Hypothesis-driven problem structuring · Translating technical findings for non-technical audiences
- **Languages:** English · Mandarin · French
- **Hobbies:** Vocals · Philosophy (Buddhist, Daoist, practical) · Local food hunter · Gym